

DS3040 Introduction to Deep Learning Lab

Assignment-1, Date: Jan 6, 2025

Timing: 10:00 to 11:00 AM

Max Marks: 20

Instructions

1. Submit one .ipynb file containing all answers. The name should be [student_name]_assignment[number].ipynb
2. Write the questions in separate text blocks before the answers.
3. Outputs for all sub-questions should be given and the code should be executable.
4. Write justifications for your choices where needed.
5. Ensure that all plots include clear labels and legends for better interpretation.
6. Do not use AI or any websites except library documentations.

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1. Use the Breast Cancer dataset from *sklearn.datasets*. This dataset contains features about tumors and a target column indicating whether the tumor is malignant or benign. (5)
 - a Load the data and split the dataset into training and testing sets (80-20 split) maintaining class proportions.
 - b Perform k-fold cross-validation (k=5) on the training set using logistic regression. Plot the accuracy scores for each fold from the k-fold cross-validation
 - c Train the logistic regression model on the full training set and evaluate its performance on the test set using the appropriate metrics.
 2. Write a recursive function named **determinant** that takes a square matrix of any size as input and calculates its determinant without using any library functions that directly compute the determinant. Compare your results with the *numpy.linalg.det()* function from NumPy. (Hint: Use *numpy.delete()* to remove specific rows or columns from the matrix.) (5)
 3. Implement a Python function to calculate the derivative of a function using the limit definition of the derivative. Plot the function and the derivative for a given range. Test the code with two functions. Using differentiation function from any library is not allowed. (5)
 4. Implement the gradient descent algorithm in Python for the following polynomial cost function: (5)

$$f(x) = 3x^4 - 6x^3 + 12x^2 - 24x + 2$$

Your output should include the optimized parameter and the value of the cost function at this optimized parameter. Also, plot the learning curve (i.e., cost vs. number of epochs).